

Behrooz Vedadian

AI RESEARCHER | DEEP LEARNING THEORY & ARCHITECTURES

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Profile

Mathematics enthusiast with a strong interest in the theoretical foundations of deep learning and learning theory. Also an experienced engineer with substantial research history in machine learning and neural architectures.

Research Interests

Applied Mathematics, Foundations of Deep Learning, Model and Data Distribution Symmetries

Education

Ph.D. in Artificial Intelligence **Amirkabir University of Technology** 2013 – 2017

- Supervisors: *Dr. Mohammad Rahmati, Dr. Shahram Khadivi*
- Research direction: *Automatic Machine Translation and graph-based neural modeling*
- Coursework and research completed

M.Sc. in Communication Systems **Amirkabir University of Technology** 2005 – 2008

- Supervisors: *Dr. Hamidreza Amindavar*
- Thesis: *Signal Detection in Nonstationary Environments using Wavelets*

B.Sc. in Electrical Engineering **Sharif University of Technology** 2000 – 2005

- Supervisors: *Dr. Mahdi Nayebi*
- Thesis: *Speaker Identification using Hidden Markov Models*

Diploma in Math. & Physics **National Organization for Development of Exceptional Talents (NODET)** 1993 – 2000

Research & Professional Experience

Targoman Intelligent Processing Co. **Researcher, Developer & Co-founder** 2015 – 2024

- Neural machine translation model training and deployment.
- Model Generalization and its relation to information bottleneck principle.
- Knowledge distillation and model compression for deployment.
- PDF Document Layout Analysis.

Projects **Researcher & Developer**

- Yara Group: Defect detection in Dental Xray images using deep learning 2021.
- Faraadid Co.: License plate recognition, Surveillance video analysis and object tracking 2017.

HLT Lab, Amirkabir University of Technology **Researcher** 2013 – 2015

- Statistical machine translation model training and deployment.
- Multifactor phrase-based SMT models and their application in low-resource language pairs.

Hoda Systems Co. **Researcher & Developer** 2005 – 2013

- Optical Character Recognition and Document Layout Analysis.
- Signature Identification.

Research Artifacts

- Graph Signal Processing for Machine Translation – 2021, *Not published*
[Link to PDF version on Github](#)
- **Graph Signal Processing-Based Neural Architecture for Sequence Modeling** – *Late 2015*
Prepared as my PhD research proposal with extended conceptual and mathematical details
Akin to encoder layers in *Transformer* architecture
Link to PDF version on Github [in Farsi](#) and [in English](#)
- Design and evaluation documentation of large-scale Persian neural MT systems – 2016.
- Design and evaluation documentation of large-scale Persian statistical MT systems – 2014.

Awards

- Ranked 9th in National Ph.D. Entrance Examination (2013).
- 3rd and 2nd place in National Computer Programming Contests (1997, 1998).
- Qualified for second round of National Mathematics and Physics Olympiads (1999).

Technical Skills

- **Programming:** C++, Python, TypeScript
- **ML/DL:** PyTorch, TensorFlow, HuggingFace Transformers
- **Mathematics:** Linear algebra, Optimization, Modern probability, Matrix calculus